

AMERICAN

Horological Journal.

VOL. I.

NEW YORK, JULY, 1869.

No. 1.

CONTENTS.

ASTRONOMY IN ITS RELATIONS TO HOROLOGY, . . .	5
WATCH AND CHRONOMETER JEWELLING, . . .	11
HINTS ON CLOCKS AND CLOCK MAKING, . . .	15
NOTICES OF NEW TOOLS, . . .	17
GREENWICH OBSERVATORY, . . .	17
PINIONS, . . .	20
NEW THREE-PIN ESCAPEMENT, . . .	23
ENGLISH OPINION OF AMERICAN WATCH MANUFACTURE, . . .	23
CORRESPONDENCE, . . .	24
ECLIPSE OF THE SUN, . . .	25
DIAMOND CUTTING, . . .	25
ALLOYS OF ALUMINUM WITH COPPER, . . .	25
EQUATION OF TIME TABLE, . . .	28

*** Address all communications for HOROLOGICAL JOURNAL to G. B. MILLER, P. O. Box 6715, New York City. Publication Office 229 Broadway.*

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Title: American Horological Journal, Vol. I, No. 1, July 1869: Devoted to Pratical Horology

Editor: George B. Miller

Release date: April 17, 2022 [eBook #67859]

Most recently updated: October 18, 2024

Language: English

Original publication: United States: G. B. Miller, 1873

Other information and formats: www.gutenberg.org/ebooks/67859

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Even with such a standard as the civil year of 365d. 5h. 48m. 49.7s., the incommensurability that exists between the length of the day and the real place of the sun makes it very difficult to adjust the ratio of both in whole numbers. Were we to return to the point in the earth's orbit in exactly 365 days, we would have precisely the same number of days in each year, and the sun would be at the same point on the ecliptic at the same second at the beginning and end of the year. There is, however, a fraction of a day, so that a solar year and civil are not of equal duration.

It is thus we have our bissextile year, from the fact that the inequality amounts to nearly a quarter of a day, so that in four years we have a whole day's gain; but not exactly, because a fraction still remains to be accounted for. Now, if we should suppress the one day of leap-year once at the end of each three out of four centuries, the civil would be within a very small fraction equal to the solar year, as given by observation; this small fraction would be almost entirely eliminated, provided we suppressed the bissextile at the end of every four thousand years. Were this fraction neglected, the beginning of the new civil year would precede the tropical by just that much, so that in the course of 1507 years the whole day's difference would obtain.

The Egyptian year was dated from the heliacal rising of the star Sirius; it contained only 365 days. By easy computation it can be shown that in every 1461 years a whole year was lost; this cycle was called the Sothaic period, in which the heliacal rising of Sirius passed through the whole year and took place again on the same day. The commencement of that cycle took

place 1322 years before Christ. The year by the Roman calendar was dated by Julius Cæsar the 1st of January, that being the day of the new moon immediately following the winter solstice in the 707th year of Rome. Christ's nativity is dated on the 25th of December, in Cæsar's 45th year, and the 46th year of the Julian calendar is assumed to be the 1st year of our era. The preceding year is designated by chronologists the 1st year before Christ, the dates thence running backward the same as they run forward subsequent to that period.

Astronomically, that year is registered 0; the astronomical year begins at noon on the 31st of December, and the date of any observation expresses the number of days and hours which have actually elapsed since that time, the 31st of December—Year 0.

The year is divided into months by old and almost universal consent, but the period of seven days is by far the most permanent division of a rotation of the earth around the sun. It was the division long before the historic period. The Brahmins in India used it with the same denominations as at the present day the Jews, Arabs, Egyptians, and Assyrians. "It has survived the fall of empires, and has existed among all successive generations, a proof of their common origin."

Nothing can be more interesting in the study of astronomy than its chronological value. La Place says: "Whole nations have been swept from the earth, with their languages, arts, and sciences, leaving but confused masses of ruins to mark the place where mighty cities stood; their history, with but the exception of a few doubtful traditions, has perished; but the perfection of their astronomical observations marks their high antiquity, fixes the periods of their existence, and proves that even at that early time they must have made considerable progress in science."

The earth revolving around the sun in an ellipse, the position of the major axis of the orbit would indicate something in regard to eras in astronomy extending not only beyond the historical period, but so far back in the past that imagination is almost at fault. The position of the major axis of the orbit depends on the direct motion of the perigee and the precession of the equinoxes conjointly, the annual motions respectively being $11''.8$ and $50''.1$, the two combined motions being $61''.9$ annually. A tropical revolution is made in 209.84 years. This being a constant quantity, we may ascertain

when the line of the major axis coincided with the line of the equinoxes. This occurrence took place about 4,000 or 4,090 years before the year 0. In the year 6,483 the major axis will again coincide with the line of the equinoxes, but then the solar perigee will coincide with the vernal equinox. So, it will be seen that the period of revolution is 20,966 years. But in the progress of this revolution there must have been a time when the major axis was perpendicular to the line of the equinoxes. A simple calculation will show that the eventful year was 1250; and so important is this event considered, that La Place, the immortal author of the *Mécanique Céleste*, proposed to make the vernal equinox of this year the initial day of the year 1 of our era. Again, at the solstices the sun is at the greatest distance from the equator; consequently the declination of the sun is equal to the obliquity of the ecliptic. The length of a shadow cast at noonday from the stile of an ordinary sun-dial would accurately determine the precise time on which this position occurs.

Though wanting in accuracy, such a measurement is of interest, from the fact that there are recorded observations of this kind that were taken in the city of Layang, in China, 1100 years before our present era is dated. This observation gives the zenith distance of the sun at the moment of the observation. Half the sum of the zenith distances gives the latitude, and half their difference gives the obliquity of the ecliptic at the period. Now the law of the variation of the ecliptic is well known, and modern computation has verified both the moment of taking the observation and the latitude of the place. Eclipses were the foundation of the whole of Chinese chronology, and recorded observations prove the civilization of that strange race for 4700 years.

Horology, with astronomy, was not neglected even as early as 3102 years before Christ, as the following will show.

The cycles of Jupiter and Saturn are very unequal, the latter being a period of 918 years; the mean motion of the two planets was determined by the Indians in that part of the respective orbits where Saturn's motion was the slowest and Jupiter's the most rapid. This observed event must have been 3102 years before, and 1491 after the year 0; but the record shows that the observation was taken before the last-named date.

Since both solar and sidereal time is estimated from the passage of the sun and the equinoctial point across the meridian of the place of observation, the time will vary in different places by as much as the passage precedes each. It being obvious that when the sun is in the meridian at any one place, it is midnight at a point on the earth's surface diametrically opposite; so an observation taken at different places at the same moment of absolute time, will be recorded as having happened at different times. Therefore when a comparison of these different observations is to be made, it becomes necessary to reduce them by computation to what the result would have been had they been taken under the same meridian at the same moment of absolute time. Sir John Herschel proposed to employ mean equinoctial time, which is the same for all the world. It is the time elapsed from the moment the mean sun enters the mean vernal equinox, and is reckoned in mean solar days and parts of days. This difference in time is really the angular motion of the earth, and by measuring it the longitude of any place on the surface of the earth can be determined, provided we have a standard point of departure, and an instrument capable of accurately dividing the time into small quantities during its transit from the meridian on which it was rated.

As will be hereafter shown, the axis of the earth's rotation is invariable. Were the position of the major axis of the earth's orbit as immutable, an observation of any star on the meridian taken at any place would always be the same. Again, the form of the earth has an important effect; the equatorial diameter exceeds the polar, thus giving a large excess of matter at the equator. Now the attraction of an external body not only draws another to it in its whole mass, but, as the force of attraction is inversely as the square of the distance, it follows that the attracted body would be revolved on its own centre of gravity until its major diameter was in a straight line with the attracting body.

The sun and moon are both attracting bodies for the earth; the plane of the equator is at an angle to the plane of the ecliptic of $23^{\circ} 27' 34''.69$, and the plane of the moon's orbit is inclined to it $5^{\circ} 8' 47''.9$. Now from the oblate form of the earth, the sun and moon, acting obliquely and unequally, urge the plane of the equator from its own position from east to west, thus changing the equinoctial points to the extent of $50''.41$ annually.

This action, were it not compensated by another force, would in time alter the angle of the ecliptic until the equatorial plane and the ecliptic coincided. There are few but have seen the philosophical toy called the Gyroscope. This toy, on a miniature scale, gives a fine illustration of the force brought in to correct the combined action of the sun and moon on the obliquity of the equator. The rotation of the earth is held in its own plane by its own revolution, the same as the gyroscope seems to overcome the laws of gravitation by its force of revolution.

But not only do the sun and moon disturb the plane of the ecliptic, but the action of other planets on the earth and sun is to be taken into account. A very slow variation in the position of the plane of the ecliptic, in relation to the plane of the equator, is observed from these influences. It must be remembered that a very slight deviation in the angle can and would be detected by observation with modern instruments. We do find that this attraction affects the inclination of the ecliptic to the equator of $0''.31$ annually.

This motion is entirely independent of the form of the earth. Now, if we assume that the sun and moon give the equinoctial points a retrograde motion on the ecliptic, we must deduct the influence of the planets. We may then calculate the mean disturbance by subtracting the latter from the former—the difference is settled by both theory and observation to be $50''.1$ annually. This motion of the equinoxes is called the precession of the equinoxes. Its consideration forms a very important element in the estimation of time, as the position of the various fixed stars, though so very distant, are all affected in longitude by this quantity of $50''.1$ —being an increase of longitude. Therefore, if we were to calculate the position of any given star in order to get a transit for mean time, or true time, we must take this quantity into consideration. The increase is so great that the earliest astronomers, even with their imperfect modes of observation, detected it. Hipparchus, 128 years before Christ, compared his own observations with those of Timocharis, 153 years before. He found the solution of the problem the same as Diophantus found the solution of the squares and cubes, by analysis. In the time of Hipparchus, the sun was at a point 30° in advance of its present position, for it then entered into the constellation of Aries near the vernal equinox.

At the present time the position of the equinoctial points shows a recession of the whole, $30^{\circ} 1' 40''.2$. At this rate of motion the constellations called the Signs of the Zodiac are some distance from the divisions of the ecliptic that bear their names. At the rate of $50''.1$ the whole revolution of the equinoctial points will be accomplished in 25,868 years; but this is again modified because the precession must vary in different centuries for the following reasons: the sun's motion is direct, the precession retrograde; therefore, the sun arrives at the equator sooner than he does at the same star of observation. Now, the tropical year is 365d. 5h. 48' 49''.7; and as the precession is exactly $50''.1$, we must suppose it takes some time for the sun to move through that arc. By direct observation it is found that the time required for such translation is $20' 19''.6$. By adding this amount to the tropical year we have the sidereal year of 365d. 6h. 9' 9''.6 in mean solar days. This amount of precession has been on the increase since the days of its first recorder, Hipparchus, as the augmentation amounts to no less than $0''.455$. By adding that to the known precession we find that the civil year is shorter now by $4''.21$ than in his time; but, as a great division of time, the year can be changed by this cause not more than $43''.$

The action of the moon on the accumulation of matter at the earth's equator is a source of disturbance that in very accurate observations for time should be eliminated. Thus the moon, with the conjoint action of the sun, depending on relative position, causes the pole of the equator to describe a small ellipse in the heavens with axes of $18''.5$ for the major, and $13''.674$ for the minor; the longer axis being directed to the pole of the ecliptic. This inequality has a period of 19 years,—it being equal to the revolution of the nodes of the lunar orbit. The combination of these disturbances changes, by a small quantity, the position of the polar axis of the earth in regard to the stars, but not in regard to its own surface. With so many disturbing causes, we must add that of Jupiter, whose attraction is diminishing the obliquity of the ecliptic by $0''.457$ according to M. Bessel.

The results of all these forces must affect the position of all the stars and planets as seen from our earth. Their longitudes being reckoned from the equinoxes, the precession of these points would increase the longitude; but as it affects all the stars and planets alike, it would make no real or apparent change in their relative positions. Nutation, however, affects the celestial latitudes and longitudes, as the real motion of the earth's polar axis changes

the relative positions. So great is the change that our present pole star has changed from 12° to $1^{\circ} 24'$; in regard to the celestial pole, the gradual approximation will continue until it is with $0^{\circ} 30'$, after which it will leave the pole indefinitely until in 12,934 years α Lyræ will be the pole star.

So far we have given only the causes that affect the meridian, and consequently our standard for time; but that point being established for the yearly and diurnal revolutions, it becomes necessary to find some means to divide the day into minute fractional parts, such as seconds and parts of seconds. This, it has been stated, is effected by means of an isochronous pendulum. On this instrument no comment is required but of the causes that disturb its accuracy much is needed. In 1672, at Cayenne, the astronomer Richter, while taking transits of fixed stars, found his clock lost $2' 28''$ per day. This was an error that arrested his attention, and he immediately attributed it to some variation in the length of the pendulum—due to other causes than atmospheric changes and expansion. He determined the length of a pendulum beating seconds in that latitude, which was 5° N. in South America. He found that that pendulum was shorter than one beating seconds in Paris, by $0.833+$ of an inch. Now, if the earth was a sphere, the attraction of gravitation at all places on its surface would be equal, and the oscillations of a pendulum would also be equal, + or - the disturbing effect of centrifugal force—an amount that can be easily determined. The real reason of the variation is found in the configuration of the earth.

The amount of the attraction of gravitation at any point of the earth's surface is found by the distance traversed by any body during the first second of its fall. The pendulum is a falling body, and may be by the same analysis reasoned on that pertains to the laws of gravitation; the centrifugal force is measured by any deflection from a tangent to the earth's surface in a second.

It follows that the centrifugal force at the poles, where there is the least motion, would not be equal to the force of gravitation, and at the equator must be exactly equal; but the deflection of a circle from a tangent measures the intensity of the earth's attraction, and is equal to the versed sine of the arc described during that time, the velocity of the earth's rotation being known, the value of the arc is deducible. The centrifugal force at the equator is equal to $\frac{1}{289}$ th part of the attraction of gravitation. Again, the uniformity of the earth's mass becomes an object of consideration.

Assuming that the figure of the earth is an ellipsoid of rotation, we will show the relation that form bears to the equal oscillation of a pendulum.

Taking the earth as a homogeneous mass, analysis gives us the certainty that if the intensity of gravitation at the equator be taken as unity, the increase of gravity to the poles eliminating the differences of the centrifugal force must be = to 2.5, the ratio of the centrifugal force to that of gravitation at the equator. Now, taking the 2.5 of $.346 = 1/115.2$, this then must be the total increase of gravitation. Did we know the exact amount of increase at every point, from the equator to the poles, a perfect map of the form of the earth could be produced from calculation; experiment being from physical causes totally impracticable. The following analysis, quoted from an eminent physicist, gives a very lucid idea of the reasoning:

“If the earth were a homogeneous sphere without rotation, its attraction on bodies on its surface would be everywhere equal. If it be elliptical and of variable density, the force of gravity ought to increase in intensity from the equator to the pole as *unity plus* a constant quantity multiplied into the square of the sine of the latitude. But for a spheroid in rotation the centrifugal varies by the law of mechanics, as the square of the sine of the latitude from the equator, where it is greatest, to the poles, where it is least. And as it tends to make bodies fly off the surface, it diminishes the force of gravity by a small quantity. Hence, by gravitation, which is the difference of these two forces, the fall of bodies ought to be accelerated from the equator to the poles proportionably to the square of the sine of the latitude, and the weight of the body ought to increase in that ratio.”

Assuming the above reasoning to be correct, it follows, that the rate of descent of falling bodies will be accelerated in the transition from the equator to the poles. Now, it has been before stated that the pendulum is a falling body; therefore, with the same length of pendulum, the oscillations at the pole should be faster than at the equator. Theory, in this case, is verified; for it has been proved by experiments, repeated again and again, that a pendulum oscillating 86,400 times in a mean day at the equator, will give the same number of oscillations at any other point, provided its length is made longer in the exact ratio as the square of the sine of the latitude.

The sequence to be derived from all the foregoing considerations is, that the whole decrease of gravitation from the equator to the poles is

0.005.1449, which subtracted from the $1/155.2$ gives the amount of compression of the earth to be nearly $1/285.26$. But this form of the earth would give the excess of the equatorial axis over the polar about $26\frac{1}{2}$ miles. The measurement is confirmed by Mr. Ivory in his investigations on the five principal measurements of arcs of the meridian in Peru, India, France, England, and Lapland. He found that the law required an ellipsoid of revolution whose equatorial radius should be 3,962.824 miles, and the polar 3,949.585 miles; the difference is 13.239 miles; this quantity multiplied by two gives 26.478 as the excess of one diameter over the other. Thus, by two different processes the figure of the earth has been determined; but another remains that is the result of pure analysis, derived from the nutation and precession of the equinoxes—for, as explained before, these effects are caused by the excess of matter at the earth's equator. The calculation does not lead us to certainty, but it does show the compression to be comprised between the two fractions $\frac{1}{270}$ and $\frac{1}{573}$. There is this advantage in the lunar theory, that it takes the earth as a whole, disregarding any irregularities of surface, or the local attractions that influence the pendulum—the difficulties of measuring an arc of the meridian being an obstacle to perfect accuracy.

The form of the earth has, however, a value confined not alone to those interested in horology—it furnishes us with a standard of weights and measures. In England and the United States, the pendulum is the unit of mensuration, or at least the common standard from which measurement is derived. It has been shown that, deducting the effects of nutation, the axis of the earth's rotation is always in the same plane. Now, the mass being the same constant quantity, a pendulum oscillating seconds at the Greenwich Observatory, has been adopted by the English Government as its standard of length. Oscillating in vacuo at the level of the sea, at 62° Fahr., Captain Kater found its approximate length to be 39.1393 inches; as this must be invariable under the same circumstances, it becomes a standard for all time. The French deduced their standard from the measurement of the ten-millionth part of a quadrant of the meridian passing through Formentera and Greenwich. They have also adopted the decimal system; yet it seems to prove that nothing under the sun is new, for over forty centuries ago the Chinese used the decimal system in the division of degrees, weights, and measures.

The antiquity of the pendulum is also shown by the fact that the Arabs were in the habit of dividing the time in observations, by its oscillations, when Ibn Junis, in the year one thousand, was making his astronomical researches. Before we lose sight of the influence of the form of the earth on the pendulum, it may be well to state another source of disturbance, arising from the combined influence of the earth's rotation and the fact that a body moving in its own plane seeks to maintain that plane. It will be seen from the very beautiful experiment showing the rotation of the earth, that if a body like a pendulum be suspended so as to be free in every direction, and not be influenced by the motion of the earth when set in oscillation in any plane, that that plane will preserve its line of motion, while the earth in its motion beneath the body can be seen to slowly move, as though the minute hand of a watch were made stationary while the dial revolved. The same principle is the one that maintains the spinning-top in a parallel position to the horizon, or the gyroscope in its apparently anomalous defiance of all the laws of gravitation. In the pendulum this tendency to preserve the same plane of motion becomes a cause of error—slight, it is true, but can be very easily remedied by so placing it that the plane of oscillation shall be parallel to the equator. It will be readily seen that this precaution will become more important as we recede from the equator; for if we were to suspend a pendulum at the pole in a true line with the axis of rotation, and if the plane of vibration remained constant, the earth would turn once around that plane in the diurnal period. During this time there would be a continuous torsion on the point of suspension, that would in time materially affect the accuracy of the instrument. The reasoning holds good for every latitude—degree of influence being the only difference.

Having given the action of the earth's form, mass, and rotation on the pendulum, there remain the disturbances due to expansion and contraction, owing to changes of temperature and those of atmospheric causes. The astronomical points to be observed are somewhat too fully laid down, but it must be remembered that an exact science requires the premises to be fully established before a sequence can be drawn.

As the standard of time depends on the passage of a star or the sun, or any known celestial object, at a certain time across the meridian of the place where the observation is taken, it was absolutely necessary to give the modes of calculation, together with the disturbing causes. Moreover, a full

appreciation of the indebtedness of horology to astronomy could not be obtained without a general knowledge of the change of the position of the major axis of the orbit described by the earth around the sun. Also, the difference between mean and apparent solar time was required to illustrate the use of the tables of equated time, the necessity of which will become patent when the use of the transit instrument for the establishment of time, or a fixed standard, is introduced. Also, the disturbing effects of the sun and moon collectively and relatively as to position, could not be passed, as they produce the precession of the equinoxes and the nutation of the pole—essential elements in the computation of time.

Watch and Chronometer Jewelling.

NUMBER ONE.

This whole subject is well worthy an article both in a scientific and mechanical sense, whether we consider the delicacy of the operations or the intractable character of the material operated on—for there has been no improvement in the horological trade of more importance to accuracy and durability of time-keepers.

The substitution of stone for common brass or gold bearings, was prompted by the inevitable wear of the holes from frequent cleaning, and the abrasion of the pivots, produced by the accumulation of dust with viscid oil; the pivot being cut away, or the hole opened too large. So long as the verge and cylinder were the prevailing escapements, the necessity for jewelling was not so strongly felt, except in the balance holes. The introduction of the lever escapement brought with it a better watch,—capable of more accurate time, but demanding an improved construction.

An Italian, in 1723, first introduced the practice of using stone for bearings. He not only conceived the idea, but was successful as an artisan in making his own jewels; ingenious and skilful as he was, however, he encountered obstacles almost insurmountable.

The art of cutting gems, it is true, was at that time well understood, but no one had attempted to drill a hole in a hard stone fine enough for a properly sized pivot. The watches at that time that were jewelled could boast of nothing more than the balance holes, and they were not pierced to let the pivot *through*.

It is a very difficult matter to polish a taper indentation in a stone, even with modern appliances, in consequence of the tendency to create a *tit* at the bottom,—thus throwing the balance staff out of upright. The difficulties in the then state of knowledge retarded the general introduction of stone-work

for many years. The Swiss, however, seeing the advantages derived, finally struck out the various manipulations with success. Time and experience gave more skill, and at the present time it is impossible to find a Swiss watch, even of the cheapest class, that is not jewelled in at least four holes. The English trade adopted the art later; but even then it did not become general for many years. Within a generation, only fine English levers were jewelled.

The mere substitution of a harder substance was not the only improvement; other conditions necessary to accuracy were insured. The hole could be made *round*—the material of such a character that no chemical action could be effected on the oil used for lubrication, and the vertical section of the hole could be made so as to present the least amount of frictional surface, yet still giving a perfectly polished bearing, thus avoiding the cutting of the pivot.

The whole “*modus operandi*” from the stone in the rough to the last setting up is well worth the attention of the watch repairer, and certainly that of the manufacturer.

Of the materials used in the trade, the first and most important is the diamond, used only in the time-piece as an end-stone—but at the bench all-important, as a means of making the other jewels. The diamond possesses the requisite susceptibility of polish, combined with greatest hardness of any substance known; but this adamantine quality precludes its being pierced with a through hole. Considered chemically, the diamond is pure carbon,—its different varieties differing only in structure—common charcoal, its lowest—plumbago, its intermediate grade. Another variety, called the “black diamond,” or “diamond carbon,” occurs, which is interesting as being a parallel with emery, compared with crystalline sapphire. The form of diamond most in use for mechanical manipulations, is almost always crystallized; yet it will be seen that the agglomerated form of diamond carbon plays no unimportant part in jewelling. As a jewel, no use is made of the diamond, other than as an end-stone. Marine chronometers, in which the balance will weigh from five to nine pennyweights, are almost invariably furnished with a diamond end-stone, set in steel. Yet, hard as the substance is, it is often that a pivot will cut an indentation in its face. The cause of this apparent anomaly is to be found in the structural character of the gem, and its value. The lapidary, saving in

weight as possible, does not care, in “Rose Diamonds,” to pay attention to the lines of cleavage. If the face of the stone makes a slight angle with the strata of the jewel, there occur innumerable small angles of extreme thinness—the pivot, coming in contact with any of these thin portions, may fracture it, and the fragment, becoming imbedded in the tempered steel pivot, becomes a drilling tool. In our experience we have had marine chronometers sent for repair, that have lost their rate so much as to become utterly unreliable from this cause alone—the pivot having produced an indentation of the stone, creating more friction, and thus destroying the accuracy of the instrument.

As a general rule, the rose diamonds sold for this purpose are sufficiently good for general work. In a very fine watch or chronometer the stone should be selected with reference to its polish on the face, and its parallelism in the lines of cleavage. The diamond, however, gets its great importance from being the only agent we can use in working other stones. Without it the whole art of jewelling would not be practicable. The various steps are all connected some way with diamond in its different shapes. “Bort,” the technical name for another variety, is merely fragments of the stone that have been cleaved off from a gem in process of cutting, or gems that have been cut, but found too full of flaws to become of use for ornamental jewelry purposes, the cost depending on the size, varying from \$5.50 to \$18 per carat. This “Bort” is used as turning tools—the larger pieces being selected and “set” in a brass wire and used on the lathe, in the same manner, and with the same facility, as the common graver. For tools, even the diamond is not of equal value—a pure white and crystalline in structure generally being too brittle (though hard) to endure the work. Among the workmen the “London smoke,” a clouded, brownish stone, is most prized—it possessing the twofold qualities of toughness and hardness.

Another form of “Bort” comes in the shape of a small globule, sometimes the size of a pea; it is crystallic, and when fractured generally gives very small, indeed minute pieces of a needle shape. These are carefully selected, and form the drills with which the English hole-maker perforates the jewel. These drills, when found perfect, for soundness, form, and size, are very highly prized by the workman, as the choice of another, together with the setting, will often take a vast deal of time and labor.

“Bort” is also used in the making of the laps or mills with which the jeweller reduces the stones to a condition for the lathe and subsequent processes. For this purpose such pieces as are not fit for cutting-tools, or drills, are selected. A copper disk, having been first surfaced and turned off in the lathe, is placed on a block or small anvil; each piece of stone is then separately placed on the copper, and driven in with a smart blow—care being taken that no place shall occur in the disk that does not present, in revolution, some cutting point. It would seem impossible to retain the diamond fragment, but it must be remembered that the copper, being a very ductile metal, receives the piece; the first rubbing of a hard stone then burnishes the burred edges of the indentations over every irregular face of the diamond, leaving only a cutting edge to project. The rapidity with which such a lap, well charged, will reduce the hardest stone, is somewhat marvellous. It is the first tool used in jewellery, and so important that a more detailed and explicit description of its make will be given when the process of manufacture is treated upon.

Diamond powder is equally as important as “bort,” being used in nearly every stage of jewel-making. The coarsest charges the “skives” or saws used for splitting up the stone. These skives are made of soft sheet-iron, and act on the same principle as the laps. The finer grades, in bulk, resemble very much ordinary slate-pencil dust; indeed, the latter is often used as an adulteration. This powder is not uniform in fineness, and the jewel-maker is under the necessity of separating the different grades. This is effected by a simple process called “floating off,” and is conducted as follows: A certain quantity of powder, say a carat, is put into a pint of pure sweet oil, contained in some such shallow vessel as a saucer. Depending on the fluidity of the oil, the mixture, after being thoroughly incorporated, is allowed to stand undisturbed for about an hour or an hour and a half. During this time, owing to their greater gravity, the largest particles are precipitated, leaving held in suspension a powder of nearly uniform fineness. The mixture is now carefully decanted into another similar vessel, leaving the coarse powder at the bottom of the first. This coarse deposit is denominated No. 1, and is used for skives, laps, and other rough purposes. The decanted mixture in the second vessel is allowed to remain quiescent for twelve hours, when the same operation is performed; and the third vessel now contains most of the oil, together with the finest particles of powder. The precipitate from the second decantation is the ordinary opening

powder; the finest being for polishing both the holes and outsides of jewels, and giving the final finish to the faces of pallets, roller pins, locking spring jewels, etc.

The good workman is careful to keep the powder in this condition as free as possible from any extraneous dust, and above all to preserve the different grades from any intermixture, as a small quantity of a coarser grade would destroy a finer one for all its purposes, and the process of “floating off” would have to be repeated.

The most important stone in jewellery, the diamond, becomes more of an agent of the manufacture than an object.

Properly, for jewellery the ruby and sapphire are pre-eminent; inferior only to diamond in hardness, possessing a sufficient degree of toughness, susceptible of an exquisite polish, this (for they are one and the same) stone is the favorite of the Swiss, English, and American, for all high class work—the Swiss, however, using it indiscriminately in all watches.

The ruby proper is of one color, but in its varieties of intensity may change to a very light pink. When still lighter it is ranked a sapphire, which comes in almost every possible color and shade, from ruby to a perfect transparent colorless crystal. This stone differs in degrees of hardness and capacity of working—the hardest being a greenish yellow, in the shape of pebbles, with very slightly rounded edges, difficult to work, but forming the strongest and most perfect jewel known.

It must be remembered that this description gives the value of the ruby and sapphire as a material for jewellery only. For ornamental jewelry, the value depending on color, of the most intense ruby or blue for sapphire, together with brilliancy and weight. The ruby and sapphire are formed on an aluminum base, the common emery being another form of structural arrangement, but of the same chemical constitution.

These stones possess every quality to make them the base of perfect jewellery; and still the chrysolite is equally in favor with most jewellers. It is not quite so hard, but it is more easily worked and cheaper in price, and it would be difficult to tell wherein it is inferior to either the ruby or sapphire. It has a yellowish tinge, verging to the color of the olive. As a stone for jewelry it is not fashionable, and only in Persia is it valued. There are,

however, some very strong objections to its use by the workman; it is not uniform in hardness; in polishing it will *drag*, that is, the surface will tear up in the process. Unfortunately the eye is not able to detect the fault before working, and it is found only when much preliminary time and trouble has been expended. It is susceptible, when good, of a perfect polish, and is much used in chronometer work, especially for jewelling the 4th hole, as its non-liability to fracture renders it valuable.

“Aqua Marine” is a brother to the emerald, differing from it only in intensity of color, and composed of the same constituents. These two gems are the only ones in which the rare metal, glucinum, has been detected. It is extensively used in the American and English watches, but never in the Swiss. It is soft, not much harder than quartz, but comes in large pieces, perfectly transparent, and of a color which is that pure green of sea-water, from which it takes its name, “Aqua Marine.”

The garnet in English watches plays an important part for pallets, also for roller-pins; a very soft stone, but very porous. When set in the pallet with a pointed toothed wheel, it is apt to act as a file from its porosity, cutting the end of the tooth. This may be detected in any pointed tooth lever watch, by observing the color of the back of the tooth. “Black vomit” it used to be called in the Boston factory. Most of the garnet used is an Oriental stone, the best quality coming in bead form, the holes having been pierced by the natives. The cost of piercing the stone in Europe or America would be far above its value. The Oriental is the best for Horological purposes, though Hungary and Bohemia furnish the most highly prized stones used for ornamental purposes; indeed, in some German towns the cutting and setting of the garnet is a specialty, giving employment to a large number of people. And, strange to say, the best market for their sale is the United States.

This comprises about all the stones used in watch and chronometer jewellery. Still in clock work the pallets are generally jewelled in agate, a stone not at all suited to the purpose, it having, even in the best specimens, a decided stratification that prevents an uniform surface being formed by any process. The cornelian form of the agate is not open to this objection, and makes capital bearings for knife edges of fine balances, and compass stones for centres of magnetic needles. For watch or chronometer purposes the only really useful stones are sapphire, ruby, chrysolite, and aqua marine—all possessing peculiarities that deserve some remarks, as they are of the

utmost importance to the hole maker. The sapphire is the hardest stone, next to the diamond, and yet specimens can be, and are found, so soft as to *drag* in polishing. Again, if stratified very clearly, will “fire crack” in opening the hole. The ruby is more uniform in its structure, and is more highly prized on that account; its hardness being all that is necessary, while its susceptibility of receiving a high polish is equal to that of the sapphire or chrysolite. The aqua marine is always uniform and may be polished both externally and in the hole with “tripoli,” saving something in diamond powder in the process of making. In our estimation, however, the chrysolite is the most valuable of all the stones. True, when purchased in the rough, many pieces will be found unfit for the jeweller’s purpose; but when the right quality is found, nothing can be better adapted to jewellery. Hard, it is easily wrought, taking a peculiar *unctious* polish, retaining oil in its most limpid condition for a long time.

These stones form the general stock by and from which jewels are made. The details of the various manufacturing manipulations, the tools used, also the setting in the work, together with the important item of the screws, will form the subject of the next article on Watch and Chronometer Jewelling. Not having been able to get our engraving done in time for publication, we are compelled to reserve the remainder for the next number.

Hints on Clocks and Clock-Making.

NUMBER ONE.

Twenty-five years of hard labor amidst the dust and din of machinery, with hands cramped, and fingers stiffened by the continual use of tools, and with a brain constantly occupied in ringing the changes upon wheels and levers in their almost infinite combinations,—it requires a degree of courage to undertake to write anything that can be dignified with the name of an “article,” although it does propose to treat upon a subject with which we are fairly familiar; but it is consoling to think that one is not expected to write for the pages of this practical journal with the same degree of elegance and polish that should grace the columns of a review or magazine; that we can appear here as plain, practical mechanics, and use good hard, round words to express our ideas, backed by an experience which should add some weight—and we welcome the appearance of the “American Horological Journal,” which is to serve a good purpose by bringing out the actual experience of men who have grown gray in the art and mystery of clock-making, and preserving, by means of the “art preservative of all arts,” their dearly bought knowledge and experience, for the benefit of those who in their turn shall follow them; and it will also benefit the people in general by giving information that will lead to the purchase of good and tasteful clocks for household use.

That such a journal is needed to enlighten us, is made plain by the fact that in almost every newspaper we have a vivid account of some wonderful clock “recently invented,” which may possess some merit, but they are so grossly exaggerated by some ignorant “penny-a-liner,” that we are almost led to believe in the Irishman’s marvellous “eight-day clock, that actually ran three weeks.” Even the proverbially correct “Scientific American,” of which I am a constant reader, has in its issue of June 19th, an account in its “editorial summary” of a clock in France containing “90,000 wheels,” and perhaps the most curious part of the mechanism is that which gives “the

additional day in leap-year,” etc. Now, it will require but little knowledge of clocks to tell us that one with 90,000 wheels was never made and never will be, but “the additional day in leap-year” has been given by calendar clocks in this country since the year 1853.

It is not proposed in the series of articles to follow, to discuss the early history of clocks. Reid and Dennison have written enough to convince the most skeptical that the clock is an old invention. It is not important to us who invented the pendulum, or this or that escapement, but who makes the best pendulum, the best escapement, the most perfect train of wheels and pinions. These are vital points, and we shall endeavor to give them that attention that their importance demands. It is proper to state here that any assertion made, or rule given, has been tested, and is the result merely of our experience, and we do not claim that it is all there is of the subject; for we are aware that the experience of others may have led to results entirely different; but if all clock-makers will avail themselves of the columns of this journal, we shall not only become better acquainted by an exchange of ideas, but better clock-makers.

The subject of wheels and pinions is of the greatest importance in clock-making, and the utmost care and skill are required to execute a train which shall not only run with as little friction as possible, but the friction must be equal; for if there is no variation in the train force, the escapement and pendulum will always be actuated by the same amount of power, and the performance of the clock can be relied upon. Clock text-books do not fully impress this subject. We find a great deal upon this or that escapement, and the different pendulums. Dennison has a couple of pages full of abstruse calculations upon a method of shifting an extra weight upon a rod, so that the going of a clock can be varied one second per day; but if his wheels and pinions are not perfect, a large tooth here and there will vary the clock more than that.

Reid overawes us with his knowledge of the proper curves of the teeth of wheels; but it must have been only theory, for his practice was to saw his teeth, and his cycloids, epicycloids, and hypocycloids were left to the mercy of the “topping file” in the hands of his “wheel teeth finishers,” instead of shaping up the teeth in the engine, as is done now. We have generally cut the wheels of fine clocks over several times with different cutters before taking them from the engine; the last cutter having but one tooth, which can

be made perfect as to cut and shape, and, running with great speed, will leave the teeth the proper shape, very smooth, and as true as the dial of the engine. Escape wheels, especially, require great care in cutting, as the teeth for dead-beat escapements are somewhat long and thin; the least inaccuracy is certain to cause trouble. It is absolutely necessary that the dial plate of the cutting engine should be perfectly true, with clean, round holes, and a perfect fitting index point, with a cutter arbor without end play or lateral motion—these are the essentials of a good cutting engine, without which a good clock cannot be made.

We have generally made a practice, upon the completion of the train for a fine clock, to put in the place of the escape-wheel a very light, well-balanced fly, to prevent “backlash,” and a very fine soft cord on the barrel; then hang on a very light weight; so slight that—all of the wheels being balanced, and no oil upon the pivots—the fly will move so slowly that its revolutions may be counted. By taking care that the weight be not too much in excess of the resistance, the least inaccuracy in the wheels and pinions may be discovered by the difference in the velocity of the fly, or by its suddenly stopping, which will be occasioned by any inequality in the train teeth, which would not have been discovered by the closest scrutiny. It was by means of this test that we discovered an inaccuracy in a pinion, caused by hardening, which could not have been discovered by a less delicate test.

The wheels in the train should be as light as possible, for as the whole train is stopped every time a tooth drops on the pallets, it is plain that the driving weight must overcome the inertia as well as the friction of the train at every beat. To this end it has been customary to “arm out” the wheels, leaving a very light rim supported by light arms, the wheels being generally of cast brass, turned up, and cut, then lightened. We followed this plan for some time, but abandoned it, as we found great difficulty in making a perfectly round wheel. The arms serve as posts to support the rim in cutting or turning, but the space between is very apt to spring down. We prefer making the wheels of fine hard-rolled sheet brass; it is superior to cast brass, much finer, harder, and more durable, and is freer from flaws. After the wheels are cut, they are turned out on each side, leaving a thin web in the centre; they can be made lighter, finished easier, and are round.

As to the shape of the teeth in clock-wheels, the subject has been so ably treated by Reid, Dennison, and Prof. Willis (who has invented an

instrument to assist in laying out the curves for the teeth of wheels), that we shall not attempt it in this paper; besides, there is so little of the entire theory that can be applied to a clock-wheel of two and a half inches in diameter, with 120 to 140 teeth, farther than to leave the wheel and pinion of the proper diameter, that we consider it unnecessary; for if makers of regulators and other fine clocks will use pinions of 16 or 20 teeth, the friction or driving is all after the line of centres, and the whole subject of cycloids, epicycloids, and hypocycloids is reduced to a very small point, and might be said to “vanish into thin air.”

Having given only a few practical hints, and not yet crossed the threshold of the subject, we propose to continue from month to month—if the readers of the JOURNAL do not weary—the discussion of the various parts that go to make the sum total of a fine clock, with notices of the various clocks made in this country.

It certainly comes within the province, and is the duty, of a journal devoted to Horology, to make a note of any and all the new improvements that pertain to the science. We give, then, some few, the merits of which have struck us as being a very important matter of consideration.

The best clock time-keeper is not absolutely perfect, so its rate must be kept; but the watchmaker ordinarily has no means of correcting the error of his regulator, until the accumulation renders it a serious inconvenience. Did he possess a Transit instrument, properly set and adjusted for meridian, together with the required books and knowledge of observing, he could from day to day correct his clock and keep accurate time; but these are all expensive, as well as involving time and labor. Suited to the wants of the artisan is a little instrument called the Dipleidescope; simple in its construction, and not liable to get out of position or order, it forms the best substitute for the transit we have seen. It is founded on the theory that the double reflection from the two surfaces of planes at an angle of 60° will coincide when the object reflected is in a true line with half the base of the whole triangle. Having a prism cut in an equilateral triangle, one angle is set directly down toward the centre of the earth, the base being brought parallel with the line of the horizon. Now, if the axis of the prism is in a line with

the meridian, a reflection of the sun will appear, at the instant of crossing the meridian, on itself—that is, there would be but one image. If the instrument is well made, there can be no doubt of its accuracy and value to those who, wishing to verify their time, are not situated so as to use a transit.

Another improvement is a Bench-Key for watchmaker's use. No one who has had any experience at the bench but will appreciate an article that facilitates the setting of time-pieces for his customers. In winding, it is equally valuable. It is not dependent for its strength of torsion on the spring-chuck principle, the power being applied close to the square by means of a pin that passes through the key.

Hall's Patent Cutting Nippers are a positive desideratum; a large wire can be cut off without the least jar to the hand, the leverage is so great. The smallest sizes are suitable to the ordinary run of watch-work, and can be used in clock-work better than any cutting-plyers extant. Strong and durable, they possess one quality that all watchmakers will appreciate—if a cutting-jaw is broken it can be replaced by another.

Greenwich Observatory.

About two hundred years ago, England began to take a lead in the mercantile commerce of the world; her ships were daily passing across the Atlantic, and India also was beginning to attract her attention. It was therefore of the utmost importance that navigators should be enabled to find their longitude when at sea, independently of watches or clocks; and a reward was offered to any one who should discover a method by which this result might be obtained.

The plan proposed was, that the angular distance of the moon from certain stars should be calculated beforehand, and published, so that, for example, it might be stated that at ten minutes and five seconds past nine on such a day, the moon should be distant from Mars 40 degrees. If from a ship in the middle of the Atlantic, Mars and the moon were found to be 40 degrees apart, then it would be known that the time in England was ten minutes and five seconds past nine.

Here, then, was one item ascertained, and the method was a good one; but in consequence of the want of accuracy as regarded the moon's motions, and the exact positions of the stars, it could not be practically carried out.

Under these circumstances, Charles II. decided that a national observatory should be built, and an astronomer appointed; and a site was at once selected for the building. Wren, the architect, selected Greenwich Park as the most suitable locality, because from thence vessels passing up and down the Thames might see the time-signals, and also because there was a commanding view north and south from the hill selected for the site. The observatory was completed in 1676, and Flamsteed, the chief astronomer, immediately commenced his observations, but with very imperfect instruments of his own. During thirty years, Flamsteed labored indefatigably, and formed a valuable catalogue of stars, and made a vast collection of lunar observations. He was succeeded by Halley, who carried on similar observations; and from that time to the present, Greenwich Observatory has been our head-quarters for astronomical observations.

The work carried on at Greenwich is entirely practical, and consists in forming a catalogue of stars and planets, and so watching them that every change in their movements is at once discovered. Now that this work has been performed for several years, the movements of the principal celestial bodies have been so accurately determined, that the *Nautical Almanac*—the official guide on these subjects—is published four years in advance, and thus we find that on a particular night in 1868, the moon will be at a certain angular distance from a star, and the second satellite of Jupiter will disappear at a particular instant. On the exterior wall of the observatory there is a large electric clock, which, being placed in “contact” with the various other clocks in the observatory, indicates exact Greenwich time. The face of this clock shows twenty-four hours, so that it requires that a novice should look at it twice before comparing his watch. On the left of this clock are metal bars let into the wall, each of which represents the length of a standard measure, such as a yard, foot, etc. And let us here say a few words about these standards. To the uninitiated a yard is simply three feet, and a foot is twelve inches—an inch being, we are told in our “Tables,” the length of three barleycorns. Now, as the length of a barleycorn varies considerably, it requires something more definite than this to determine our national measures. Thus, the question, what is a foot? is more difficult to answer than at first sight appears. Many years ago the French perceived the difficulty appertaining to the national standard, and they, therefore, decided that a metre should be the ten-millionth part of one-fourth of the earth’s circumference—that is, ten-millionth of the distance from the Equator to the Pole. But here another difficulty was encountered, because different calculators found this arc of different lengths. By *law*, however, it was decided that one measurement only was correct, and so the metre was fixed at 3.0794 Paris feet; though since then, more accurate observations and improved instruments have shown these measured acres to have been very incorrectly ascertained, and thus the French method failed when practically tried.

The length of a seconds pendulum oscillating in a certain latitude has been our method of obtaining a standard; but this also has its weak points, so that to obtain a constant standard it is necessary to have some pattern which is unchangeable, and thus a metal has been chosen that expands or contracts but little either with heat or cold; and this, at a certain

temperature, is *the* standard measure, and such a standard may be seen on the exterior wall of Greenwich Observatory.

On entering the doorway—which is guarded by a Greenwich pensioner, who will possibly first peep at the visitor, in order to see who the individual may be who is desirous to tread within the sacred precincts—one finds a court-yard, on the left of which are the transit-room, the computing-room, and the chronometer-room. The transit room takes its name from the instrument therein, which is a large “transit.” This consists of a large telescope, the outside of which is not unlike a heavy cannon, as it is of solid iron. The instrument is supported by trunnions, which allow the telescope to be elevated or depressed to point south or north, and, in fact, to make a complete revolution, but never to diverge from the north or south line. The magnifying power of this instrument is not very great, so that it admits plenty of light, for it is intended, not as a searcher for or for gazing at celestial objects, but for the purpose of noting the exact time at which stars and planets pass south or north of Greenwich. Upon looking through this telescope, the observer’s eye is first attracted by a vertical row of what seem to be iron bars, placed at equal distances from each other. These, however, prove to be only spiders’ webs, and are used for the purpose of taking the time of passage of a star over each wire, and thus to ascertain the exact instant of its being in the centre of the telescope. During even the finest and calmest nights, there is occasionally found a tremulousness in the instrument, which, as it is rigidly fixed to the walls of the building, must be due to a slight vibration in the ground itself. Thus, many a feeble earthquake unfelt by the outsider may be perceived by the astronomer by the aid of his delicate instruments.

The various stars seem to be travelling at an immense rate when seen in the field of the transit telescope, and it is really nervous work noting the exact time when each wire is passed. The experienced observer, however, not only will give the minute and second, but also the decimal of a second when the star was on the wire. The result is obtained by counting the beats of a clock the face of which is opposite the observer. Thus, if at three the star seems as much short of the wire as at four it had passed it, then 3.5 might be the instant of “transit.”

At noon each day the sun’s passage is observed by nearly the whole staff of observers. One individual looks through the telescope, and gives the time

for each wire, while others examine a variety of micrometers in order to ascertain the fractional parts of seconds, etc.,—these micrometers being placed at the side of the instrument.

In the morning, the principal work consists in making what are termed the “reductions” to the observations of the previous night. These reductions are the corrections requisite for the slight instrumental inaccuracy, for the refraction of the atmosphere, and for the known constant error of the observer. When, therefore, a bright winter’s night has occurred, the work on the following morning is usually very heavy. At noon the sun’s time of transit is taken, and at one o’clock the “ball” is dropped, by means of which the various vessels in the Docks and in the Thames set their chronometers, or ascertain their rate. In addition to this, the time is sent by electricity to Deal and one or two other seaports, in order that every vessel may be able to know the accurate time, if within sight of those places.

Not the least interesting portion of the observatory is the chronometer room. For a very small charge, manufacturers or owners may have their chronometers rated at Greenwich, which is accomplished in the following manner:

The chronometer is placed in the chronometer room, and compared with the large electric clock in the room, this clock being kept in order by the stars. Each day the chronometer is examined, and thus its rate is ascertained in its then temperature. It is afterwards placed in a sort of closet warmed by gas, a condition supposed to represent the tropics, and it is there kept for a certain period, being tested each day as before. This change of temperature is found to produce very little effect on the best instruments, which, when they have passed the ordeal, are returned to the owners with their character ticketed to them. Some hundred chronometers are often placed in this room; and to compare them is a science, the “expert” by a glance discovering the difference between the two instruments, whilst a novice would require to mentally add or subtract, and thus slowly to arrive at the same results.

As soon as it becomes dark enough to see stars by the aid of a telescope, one of the staff commences his observations. These are continued during the night; and a register is kept of each star, planet, comet or moon, which is “doctored” in the morning by the computers.

As all mortals are fallible, it is desirable to bring machinery into use where possible, and this has been managed in connection with astronomical observations. Instead of the computer registering by judgment the time of a star's transit over the various wires, he strikes a small indicator, which, completing the electric circuit, causes a pricker to fall and make a hole in a piece of paper that is attached to a slowly revolving barrel. Each time the star passes a wire, the pricker descends and leaves its mark; and the interval between these marks being measured by scale, the mean time of transit may be obtained.

There is usually a feeling of the sublime that comes over us when we reflect upon the vast unexplored regions of space, or contemplate the stellar world that shines upon us. The magnitude and grandeur of some of the planets in the solar system strike us with a feeling of awe and wonder, while we are puzzled at the mysteries attending comets, double stars, nebulae, etc. No such feelings or sentiments, however, are allowed to enter into the constitution or mind of an observer at Greenwich. Saturn, the glorious ringed planet, with its galaxy of moons, is simply "Saturn, Right Ascension 10 hours 8 min. 12 sec., North declination $16^{\circ} 12' 2''$." Anything appertaining to the physical constitution, the probable cause of the ring, or the object of so grand an orb, does not come within the range of the observations at Greenwich, which are limited to bare matter-of-fact business work.

The southern portion of the observatory ground is devoted to the investigation of meteorological subjects, and is under the superintendence of Mr. Glaisher, who is now well known as an aerial voyager. It is here that an exact record is kept of the amount of rain that daily falls, of the direction and force of the wind, of the magnetic changes, of the temperature, amount of ozone, etc.—all matters which may, and probably will, lead us eventually to the discovery of some laws connected with the states of weather, and enable us to predict what may be expected from day to day. Whilst we are now able to calculate to a few seconds, and for years in advance, the instant when an eclipse may occur, and to explain the causes of the various planetary movements, yet we are in a sad state of ignorance as regards the causes of hurricanes, thunder-storms, continued rains and droughts; and thus we find that all the would-be prophets who from time to time spring up and oracularly announce a coming frost or fine weather, or the reverse, are

perpetually meeting with most signal failures, which, however, does not deter future adventurers from attempting to gain a cheap temporary renown by trying their luck at a prophecy.

The perpetual accumulation of facts at Greenwich, whether these be of an astronomical nature, or appertaining to the air we breathe and its subtle changes, is a proceeding that must eventually lead us on to a correct knowledge of the laws which govern these matters, and also keep us acquainted with any variations that may be occurring in the elements that surround us.

The order and quietness necessary in such calculations as those carried on at Greenwich prevent it from being a “show” establishment, and hence visitors are not admitted except on special business. Then, however, every aid and assistance are offered to the student and inquirer; the use of books and instruments is freely given, and such information supplied as the little spare time of those belonging to the establishment enables them to afford. Thus a visit to or a period of study at Greenwich Observatory will amply repay those who wish to gain the latest and most accurate information on astronomical subjects, or to practise themselves at the adjustments and use of the instruments; and to those who have not such opportunity, we offer this slight sketch.

[Chambers' Journal.]

Pinions.

Well made as to truth of centring, of division, of form of leaves, and polish, are, as the trade well knows, of vital importance to the value of the time-piece.

The making and finishing is one of the most troublesome, as well as most expensive of all the processes in watch work. The nature of the material renders it difficult as it approaches so nearly in hardness to the tools used in cutting. In the ordinary Yankee clock, the *lantern pinion* has entirely superseded the solid leaf, which substitution was the greatest element of success in their cheap construction. The lantern pinion is really a nearer approximation to the required anti-frictional form than a majority of cut pinions in ordinary clocks. In the process of manufacture of the cut variety, the first consideration is the quality of the steel to be used. For this purpose it should be carefully selected by trial, thus ascertaining its fineness, uniformity, softness when annealed, together with its capacity for taking a good temper, with the least amount of springing during the hardening process. Very few pinions are cut from the solid piece—the drawn pinion wire being quite good enough, when milled and finished, for the ordinary run of watch work.

The steel wire having been selected, the first process is to cut it up in lengths a trifle larger than the required pinion. The separated pieces are then centred with care, and having been placed in a lathe, the staff and pivot are turned up to nearly the required gauge, leaving a portion of the whole piece the full size for the leaves. They are now taken to the milling tool to have the proper form given to the leaves. As this form is of the highest importance, it may be as well to give here the reasons. Supposing a wheel of 60 teeth, depthing into a pinion of 8 leaves, it can readily be seen that the arc of the motion of the wheel tooth is of greater radius than that of the leaf of the pinion, and it follows that if the teeth and the leaves are made in taper form with straight sections, there must occur a sliding motion on the surfaces of both—the power thus absorbed being totally wasted; but if we

curve the surfaces we may approach a form so nearly perfect that the wheel teeth, being motors, really roll on the leaves, avoiding almost entirely the friction caused by sliding; the necessity for this curvature becoming greater the more the wheel exceeds the pinion in diameter. This curve, which has been demonstrated by very profound mathematical researches, is the “epicycloidal;” theoretically it should give no more sliding motion than the surfaces of two plain wheels revolving on each other. To obtain this perfect form, very great pains have been taken and expenses incurred, especially by the makers of the best time-keepers.

In the American factories the cutters are very elaborately made, the section being an object of great solicitude—it being an exact counterpart of the space between any two leaves, and also of one-half the top of the leaf from the curvature to the point, so that in milling, the space made by the cutter is its shape, leaving the leaf of the proper form. Generally the pinion passes under two cutters; the first to strike down the rough stock, the other to dress it to size and shape, with a light cut. The care and skill required to make these is certainly very great, and it is a proof of the wonderful ingenuity of man that they are made so perfect as to shape and cutting power.

A very ingenious device is used for dividing the leaves under the cutter, which revolves at a moderate speed over a slide, carrying a pair of centres, between which the turned up piece of pinion wire is placed. The slide is now pushed up to and under the cutter, and in its passage as much of a cut is taken as is desirable; in drawing back the slide the fresh cut space passes under a flat piece of thin steel, screwed on the frame, and set at a slight angle to the axis of the centres. On moving the slide towards the cutter for a fresh cut, the steel plate takes the last cut, and in passing by it the pinion is turned just as much as the angularity of the plate, which must be just one leaf. By this very clever device the division is effected without an index plate. This process, however, is not good enough for work intended to be very accurate—the pinion wire not being always, or indeed rarely correctly divided, the original error will be perpetuated in all the subsequent processes. These are all milled, with oil or soda water for a lubricator, and it follows that the speed of the cutter is regulated to get the greatest cut without dulling the tool. When dull, however, the mill is sharpened on the *face* of the cutting tooth by means of small grinders of iron, using Arkansas

oil-stone dust for the first grinding, and giving the necessary delicacy of the edge by means of crocus, or sharp, followed, when fine work is needed, by rouge.

It is necessary that this care should be taken, for if the edge is left coarse it will become speedily dulled, and leave a very unequal and rough surface on the cut of the pinion, which in the subsequent grinding gives rise to error in shape and size. The pinions, thus cut to gauge, are dried in sawdust, hardened, and tempered; the staff and pivots are now turned up to size, and then pass to the polishers. In the factory they are finished by means of what are called *Wig-Wags*, which it may be interesting to the reader to have a general description of.

Two Vs are arranged as centres, the pinion is placed between them, the circular parts resting in each V, but free to turn on its own axis. Immediately above the Vs is a frame on which a slide, carrying the polisher, may traverse—generally about two inches. This slide is movable vertically so as to accommodate itself to the pinion; attached to the slide is a connection which leads to a vertical lever, which is put in motion from a crank on the counter shaft. The grinding is effected by bringing the grinder, charged with oil-stone dust in oil, in one of the spaces of the pinion, which, of course, is so arranged as to bring it parallel and central with the grinder. The power being applied, the slide takes a very rapid reciprocatory motion, and the face of the grinder, so charged, rapidly reduces the uneven surface left by the cutter to what is called the *gray*.

The form of this grinder must be as perfect as the cutters, and the care taken to get the requisite parallelism is in equal proportion, and in all the best polishers is planed up while in its position. The grinder is composed of tin and lead, with sometimes a slight admixture of antimony, rolled to an even thickness, cut off in suitable lengths, and then mounted in the carrier of the Wig-Wag to be planed up to shape. There are too many minute adjustments in the machine to render a full description in this article admissible. It is large compared with the work it has to perform, but it is very admirably made, as indeed all the tools are, in the American factories.

The polishing of the leaves is the next step, and this is effected by means precisely the same as grinding. In each stage the pinions are thoroughly cleansed before entering on another. The polisher is made precisely like the

grinder; but instead of oil-stone dust, crocus mixed with oil is substituted. Owing to the less cutting quality of the material used, the polisher loses its form sooner than the grinder, and has to be more frequently reshaped. In very fine work the crocus is succeeded by fine well-levigated rouge to bring up that jet black polish, which is considered a mark of quality by chronometer and watch makers.

With the exception of turning up the staff and pivots, all the work hitherto described has been expended on the leaves—a very tedious process, yet done, when the tools and materials are in proper order, with marvellous rapidity; but tedious as these have been, there are two others quite as much so before the leaves are finished.

The ends are to be faced—they must be flat (that is a true plane) and receive the same finish that the leaves took, and is effected by the wig-wag; only the pinion revolves between centres, at a high speed, the grinder being brought up to the turned face. Two motions operate—one rectilinear, the other circular—the result being a compound motion which prevents the grinder from touching the same spot twice in succession. To effect this more surely, the operator gives the grinder a slight vibratory vertical motion. The polishing of the two faces is effected in the same manner as the grinding; in all cases the cutting face of the grinders and polishers being kept in a plane perpendicular to the axis of the pinion, both vertical and horizontal.

The staff and pivots being in the same condition they came from the lathe, the next step is to grind and polish them. Before, however, we treat on this process, it may not be amiss to give the general watch repairer a process by which the facing may be done on a small scale.

As a rule, when the watch repairer has to replace a pinion he selects one from the material dealer, finished in the leaves, but not on the ends or faces. The following operations are simple, and any one may finish these faces with little trouble. Having turned up your pivots and squared down the face of the leaves with the turning tool, grind it in the lathe by means of a ring of metal, the inside diameter being somewhat larger than the diameter of the staff. This ring is held between two centres, thus allowing it a vibratory motion, so that when it comes up to the face it accommodates itself to its plane, and thus has no tendency to force it out of a true flat; the ring, being larger than the staff or pivot, admits a small lateral motion, enough to effect

a continuous change of surface. The same little tool may be used for polishing by substituting another polisher and using crocus and rouge. For the repairer, perhaps on general work the rouge would be superfluous. Vienna lime, used with a little slip of boxwood, brings up a very fine and brilliant polish, and in replacing new work in an injured time-piece, the steel may always be polished with great rapidity by using the lime on the gray surface left from the oil-stone dust; being quickly done and affording a very handsome finish.

To resume the consideration of the pinion, the last stage is the polishing of the circular portions. Here again the wig-wag is the most useful tool, but it operates somewhat differently, for the grinder or polisher is pressed down by the finger of the operator, the pinion being held between the centres of a small lathe attached to the wig-wag; the staff is first ground and polished as the leaves have been before, and this is the last operation performed with the pinion between centres. From this stage it is chucked in a lathe very peculiarly fitted, the mandrel being hollow; and in it is fitted what is called a pump-centre, which is movable in direction of the axis of the mandrel, and capable of being securely fastened at any desired point. On the nose of the mandrel is secured a hollow steel chuck, the two sides of which have been filed out, thus leaving an open space between the end of the pump-centre and the end of the chuck. On this end a small steel plate, extremely thin, is fastened by means of shellac, and a hole drilled in the plate capable of taking in the chamfer on the shoulder of the pivot. The pump-centre being drawn back, the pinion is introduced into the chuck, the pivot placed in the hole in the steel plate, and the pump centre is drawn forward until it forces the chamfer to fill the hole; the pivot projecting from the chuck is now ready for all the grinding and polishing processes. Here the wig-wag steps in again, and from the delicacy of the pivots is modified to suit the case; this is done by having a polisher hung in the wig-wag on centres, so it may revolve; when in operation one side of the polisher rests on the pivot, the other on a ruby placed in a screw, and which screw enables the operative to insure the parallelism of the pivot. The ends of the pivots are next rounded off and finished in another set of tools. The pinion is now ready for use, assuming it to be of the proper gauge. In the American watches the scape and fourth wheels are generally staked on the staff pinch tight; the third and centre are staked on the pinion leaves, a rebate having been turned down on the ends, the wheel set on the shoulder, and the

projecting ends of the leaves riveted down. This has not been designed as an exhaustive article on pinions; it is merely intended to open the subject as pursued in the factories. There is much more to be said; and the various processes on the small scale, as performed by the Swiss and English, together with their tools, will bear more than a general description, as they are applicable at any watch bench.

The subject will be continued, in the effort to give a full and useful article.

New Three-Pin Escapement.

A contributor to the *London Horological Journal* gives the following description of his invention:

“The merit of this escapement is in a newly invented escape-wheel which is self-locking and requires no banking pins; the pallets are curved inside the impulse and outside the locking, to work with the curved points of the teeth of the wheel; being made of gold the wheel will go without oil. From its form it has the power of double impulse and double locking with the lever. The first takes place at the discharge of the escapement, the second does not act unless the watch receives a sudden motion, and then the pin or pallet in the roller strikes lightly on the lever, when the propellant power drives it back again. The balance passes through two turns before the second locking takes place, and is formed so as to be able to take up the lever, and the watch soon rights itself, and its time will not be affected. Another advantage is, that the lever is made of a flat piece of steel, as I have introduced a gold stud to receive the ruby impulse stone, which is made to adjust easily so as to bring the escapement to the closest geometrical accuracy. By its formation this ruby guides the impulse to the external edge of the roller notch. These advantages, and its simplicity, render it suitable to the best chronometer watches.”

A FEW years ago, in 1859 or '60, Mr. Peabody, a very talented gentleman of this city, patented a three-pin escapement that performed extremely well. A full description of his patent and plan is not at hand, but we will endeavor to give it to our readers in our next issue.

English Opinion of American Watch Manufacture.

In the London circle of Horologists, more attention is paid to the scientific departments than the mercantile; but for all that, a Mr. Henry Ganney has held forth before the “British Horological Institute,” on “American Watch Manufacture.” Though an Englishman, with English prejudices, he certainly gives a very fair and impartial statement of the subject; yet he views it almost entirely in the money-making aspect. He gives all the credit deserved to American enterprise and ingenuity, and yet there is a certain sense of a drawback. He had before him samples of machine work; among others, to quote, “several movements made by the British Watch Company, which flourished and failed about twenty-five years ago; these were machine-made, and the perfection and completeness of the machinery they used for producing these frames has not been equalled, I believe, in America; several machines being used there to accomplish what was begun and completed by one here.”

Mr. Ganney is right in his statement, but the example given by the British Watch Company was the rock seen by the American navigators. One tool, for facing off, truing up, drilling, depthing, and doing all the work on the pillar plate, having cost, before completion, some three thousand pounds sterling, and from its very complexity being utterly inefficient—worse than useless. In the very inception of the American watch manufacture a similar mistake was almost made. Experience and sound reasoning proved, however, that a multiplicity of operations in any one machine rendered it entirely too complex, the adjustments too numerous, and the work totally worthless. We shall in another number refer again to Mr. Ganney’s lecture, and perhaps give some beamings of light on the early history of the American watch manufacture, derived from personal observation at the time.

Correspondence.

EDITORS HOROLOGICAL JOURNAL:

I received a Prospectus a few days ago advising me of your contemplated existence. I could hardly believe the fact; “the news was too good to be true.” However, I shall take it for granted, for I cannot see why somebody has not before had the enterprise to launch out in the periodical line on subjects connected with Horology, the field being so extensive and the want so severely felt. Enclosed I send you the subscription price; in this much I have accepted your invitation, but I also enclose some few lines on a subject not particularly practical or theoretical, but very near the truth, and may perhaps give you a view of our wants.

To tell the “plain unvarnished truth,” I am a watch repairer, located in a small country village, with a decent stock of tools and a moderate trade. In all this I am no exception; so I write this in the name of all who are similarly situated. Isolated as we are, we (the country village watch repairers) have few means to improve our knowledge of the trade, but work on the same old principles learned when we were boys and apprentices, and of better and more expeditious ways of doing our work we are entirely oblivious. True, our friends of the Hebraic persuasion, who, angel like, bring us face to face with the outer horological world by selling us material and tools, occasionally present to our benumbed vision something new, such as a Swiss lathe, or lathes used in the factories; but of what use are they to us? We purchase one; well, on the bench it may be an ornament, but for use, drilling large holes is the height of our ambition. We have not the time to learn by self-experience all the boasted usefulness and capacities of the tool; so we go back to our old verge or Jacot lathe when we have to put in a pivot or a new staff. We may know all about the escapement and be able to detect the cause of any trouble with it, but we have no knowledge of the

latest modes of repairing the injury when it is discovered, and this knowledge is what I hope to find in your journal. I live in a section where the general class of work is of a very low grade, even the old verge being very common. Our stock of material has to be heavy in proportion to our trade, and then once in a while we are compelled to send our work to the city, some sixty miles distant, in consequence of not being able to do it, either from a lack of the material or want of a proper tool. To all intents and purposes we remain as stationary as the oyster. Not only do we have these vexations, but the ignorance of the public at large as to the treatment of their time-keepers is a fruitful source of annoyance; we are often charged with fraudulent practices, and a certain degree of caution is observed by more than the most ignorant. Thus, a few days ago, a stalwart son of the Green Isle made his appearance in front of the counter, and, projecting in front of our optics a huge English double-cased verge watch, spoke in almost dramatic tones:

“Plase, sir, av’ ye could make me ticker here go, sir?”

Answering in the affirmative we reached for the silent “ticker.” He drew back with alarm.

“Bedad, an’ ye’ll not stale a morsle frae this?”

“Well, but let me see the watch.”

“An’ will ye let me eyes be on yes all the time?”

“Yes.”

“An’ yes’ll not stale a jewil?”

“No.”

“Thin, there it is.”

On looking at the movement the verge was found broken, the injury explained, and the price given. He decided on the repairs being done, but said, “Give me the watch now and when ye gets the thing fixed its meself will come and git it and pay yes.”

“But we cannot repair the watch without having it.”

“Faith, thin, ye’ll not have it; ye’ll be taking something frae it.”

Now, this is an extreme case of ignorance, pardonable, perhaps, in this instance, but the public embraces multitudes just as ignorant where an allowance cannot be made. I do not expect the JOURNAL to reach such cases, or to influence the general mass, but my hope is that it will, by raising the general self-respect and tone of the repairers, indirectly elevate the respect felt for them by the public at large.

But I am writing too long and rambling a letter. I wish to express my hearty wishes for your prosperity. And, in conclusion, will you allow me to express a hope that you will give us the knowledge we need—that is, post us up on the minutiae of repairing in the latest styles, the newest processes devised, and, above all, give us an article on the lathe and its uses?

Yours truly,
W. L. C.

We have the pleasure to give our correspondent the assurance that an expert will contribute to our next number an article interesting as well as valuable in instruction as to the use of the lathe.

Eclipse of the Sun.

The approaching total eclipse of the sun, on the 7th of August next, is exciting much interest. The obscuration first occurs in latitude $39^{\circ} 53' 3''$ north, longitude $138^{\circ} 37' 4''$ west—Washington being the meridian. The first totality is on the Pacific coast of Siberia, at sunrise, in lat. $52^{\circ} 41' 9''$ north, and long. $165^{\circ} 26' 4''$ west. The eclipse is total at noon in Alaska, lat. $61^{\circ} 46' 9''$ north, and long. $68^{\circ} 4' 6''$ west. The line of the total eclipse now runs south-easterly, grazing the coast near Sitka, thence north into British America; then entering the United States, near the head of Milk River, long. 30° W.; thence through the south-west corner of Minnesota, diagonally through Iowa, crosses the Mississippi at Burlington; thence through Illinois, a little north of Springfield, crosses the Ohio river at or near Louisville, Ky., passes through the south-west corner of West Virginia, through North Carolina, just south of Raleigh, ending on the Atlantic coast at sunset, just north of Beaufort, N. C., in lat. $31^{\circ} 15' 2''$ north, and long. $9^{\circ} 36' 6''$ east. The line thus described will be that of totality, only partial in any other part of the United States.

The United States Government is, or has been, establishing a meridian line at Springfield, partly to make observations on this coming eclipse, and with the further view of determining a standard of surveyed lines—all of the Government surveys in Illinois having been geodetic. Professor Austin, of the Smithsonian Institute, is in charge of the work, aided by an able corps of assistants.

Diamond-Cutting.

At the Great Exhibition in Paris, in a part of the park contiguous to the Netherland section, M. Coster, of Amsterdam, has erected a building wherein all the processes of diamond-cutting are carried on.

The first rough shaping of the more important facets of the brilliants is here seen performed by the workman, who operates on two diamonds at once, by bruising each against the other, angle against angle. The dust that falls from the stones is preserved for the subsequent processes of grinding and polishing those facets that distinguish the many-sided brilliant from the dull, original crystal of the diamond. It is used, mingled with oil, on a flat iron disk, set revolving with vast rapidity by steam-power, the stone itself being held upon this disk or wheel by a tool to which it is attached by a mass of fusible metallic alloy, into which the stone is skilfully inserted. Skill of eye and hand, only attainable by great practice, is needed for this work; but a skill not less exact is needed for another process, which may here be seen in daily operation—the process of cleavage. The diamond, when a blow is struck on an edged tool placed parallel to one of the octahedral faces of the crystal, readily splits in that direction. But to recognize the precise direction on the complex and generally rounded form of the diamond crystal; to cut a little notch by means of a knife edge of diamonds formed of one of the slices cleaved from a crystal, and to cut that notch exactly the right spot; then to plant the steel knife that is to split the diamond precisely in the right position; finally, with a smart blow, to effect the cleavage so as to separate neither too large nor small a portion of the stone—these various steps in the process need great skill and judgment, and present to the observer the interesting spectacle which a handicraft dependent on experience of hand and eye always affords. But Mr. Coster's exhibition has other objects of interest. For the first time, we may see here, side by side, the diamond with the minerals that accompany it in the river beds of Brazil; and there are even examples in which crystals of diamonds are included within a mass of quartz crystals, which have all the appearance of having been formed simultaneously with deposits of the diamond.

The different districts of Rio and of Bahia are thus represented—the former producing a confusedly crystallized sort of diamond termed “bort,” and the latter an opaque black variety; both these kinds being found associated with the crystallized diamonds used for jewelry. Though useful in state of powder, the black carbon and “bort” are incapable of being cut as a jewel.—“*Maskelyne’s Report,*” *Great Exhibition*.

The Alloys of Aluminum with Copper.

When Sir Humphrey Davy announced the fact that soda, lime, potash, magnesia, and the other alkalies were but oxides of a metallic base, it would have been deemed chimerical to have supposed that the discoveries he made by the expensive aid of the battery would at later date become of really commercial value. He did obtain both sodium and potassium in the metallic state. The substances in this form were new to the chemical world, still more strange to the popular. So new was it to the chemists, that, on a globule of the reduced sodium being presented to a very distinguished chemist, he, with some enthusiasm, examined it; and, admitting the fact of its being a metal, exclaimed, “how heavy it is!”—when the real fact was that its specific gravity was less than water; the expression was the result of the general preconceived opinion that a high specific gravity was a test of a metallic body. It was reserved for a French chemist, Henry St. Claire Deville, to utilize the metal sodium, and that, too, in such a manner that the demand aroused attention to its production;—demand will inevitably bring a supply.

The original reduction was made by Davy, by means of the voltaic battery. After it had been proved that these bases were really metals capable of reduction, chemistry brought all its resources to bear on the problem, and they were produced by other methods than the battery. All the processes adopted, however, were too expensive and laborious, involving an extraordinary amount of complicated manipulations with but inadequate results. The metal sodium, which is the immediate subject of our inquiry, long remained an object simply of curiosity or experiment in the laboratory.

The methods of reducing the metal have of late years been so simplified that, to quote Prof. Chas. A. Joy in the *Journal of Applied Chemistry*: “A few years ago a pound of this metal could not have been purchased for two hundred dollars, and even at that price there were few manufacturers hardy enough to take the order. At the present time it can be readily manufactured

for seventy-five cents, if not for fifty cents a pound; and the probabilities are that we shall soon be able to obtain it for one-quarter of a dollar.”

Deville found that by the reaction of the metallic sodium on common chloride of aluminum a reduction was effected; the chlorine taking up the sodium, forming chloride of sodium (common salt), while the aluminum was left free in the metallic state. It is hardly necessary to go into the particulars of the process; but a metal well known to exist, had, for the first time, been brought to the world in such a condition of structure that its qualities could be tested, not only chemically, but mechanically. This was the direct result of Deville’s metallurgic process of obtaining the reducing agent—sodium.

Aluminum in itself would be of but little use, so that a brief description will be all that is necessary. It is about the color of silver, but susceptible of a higher polish, especially on a fresh-cut surface; it is much less susceptible of oxidization than silver; its specific gravity is but little more than pine wood, and its tenacity, ductility, and laminating qualities are nearly equal to silver. Its use in the mechanical arts is limited, notwithstanding all these qualities, from the fact of its low point of fusibility, and at the heat of the fusible point being easily oxidized, so much so as to prevent soldering, except by an autogenous process. But aluminum does possess a property peculiar to itself—that of forming a purely and strictly *chemical alloy* with copper. It unites with it in any proportion; the compound formed by the addition of 10 per cent. of aluminum to 90 per cent. of copper has been found to possess all the properties of an entirely new metal, with qualities that render it a very valuable material in all fine work, such as astronomical instruments; and very fine machinery, such as watch-lathes, etc.

The French reports on the alloy are somewhat voluminous, but we give the following.

The color of this bronze so closely resembles that of 18 carat gold, such as is used for the best jewelry and watch-cases, that it is capable of receiving the highest polish, and is far superior in beauty to any gilding.

Samples taken from different parts of the largest castings, when analyzed, show the most complete uniformity of composition, provided only that the two metals have originally been properly mixed while in a state of fusion.

These experiments have been made upon cylinders weighing many hundreds of pounds, and are entirely conclusive.

This valuable quality is not found in any of the more ordinary alloys of copper. The alloy of copper with tin, for example, known as *gun metal*, is notoriously subject to a phenomenon known as *liquation*; in consequence of which a great difference is found in the composition of the same casting, both in the top as compared with the bottom, and in the centre as compared with the circumference.

This phenomenon often causes great inconvenience, as the different parts of large objects will in consequence vary greatly in hardness as well as in strength. In casting artillery the difficulty becomes a serious one, and no means have yet been discovered by which it can be entirely removed.

This homogeneousness of aluminum bronze is a natural consequence of the great affinity existing between the two metals of which it is composed; and that there is such an affinity is clearly proved by the phenomenon attending the manufacture of the alloy. The copper is first melted in a crucible and the aluminum is then added to it *in ingots*. At first there is, of course, a reduction of temperature, because the aluminum in melting absorbs the heat from the melted copper; and this absorption is so great, in consequence of the great capacity for heat of aluminum, that a part of the copper may even become solid. But let the mixture be stirred a moment with an iron bar, and the two metals immediately unite; and in an instant, although the crucible may have been removed from the furnace, the temperature of the metals rises to incandescence, while the mass becomes as fluid as water.

This enormous disengagement of heat, not seen in the preparation of any other ordinary alloy, indicates, not a simple mixture, but a real chemical combination of the two metals. The 10 per cent. bronze may therefore be properly compared to a salt, the more so as it is found by calculation to contain, within a very minute fraction, four equivalents of copper to one equivalent of aluminum.

The 10 per cent. bronze may be forged cold, and becomes extremely dense under the action of the hammer. The blades of dessert-knives are thus treated in order to give them the requisite hardness and elasticity. But it has another valuable quality which is found in no other kind of brass or bronze:

it may be forged hot, as well as, if not better than the very best iron. It thus becomes harder and more rigid, and its fracture shows a grain similar to that of cast steel. On account of the hardness of the aluminum bronze, rolling it into sheets would be a tedious and expensive process, were it not for this property of being malleable at a red heat. But it may in this manner be rolled into sheets of any thickness or drawn into wire of any size. It may also be drawn into tubes of any dimension.

From several experiments made at different times at Paris, it appears that the breaking weight of the cast bronze varies from 65 to 70 kilogrammes the square millimetre. The same bronze drawn into wire supported a weight of 90 kilogrammes the square millimetre. The iron used for suspension bridges, tested in the same manner, did not show an average of more than 30 kilogrammes. Some experiments were also made by Mr. Anderson, at the Royal Arsenal at Woolwich, in England, who tested at the same time the aluminum bronze, the brass used for artillery and commonly called *gun metal*, and the cast steel made by Krupp in Prussia. Taking for the maximum strength of the bronze the lowest of the numbers found as above, we are thus enabled to form the following table of comparative tenacities:

Aluminum bronze 10 per cent.	65
Crupp's Cast Steel	53
Refined Iron	30
Brass for cannon	28

The comparative toughness of these same four metals was also tested in the following manner: A bar of each was prepared of the same size, and each bar was then notched with a chisel to precisely the same depth. The bars were broken separately, upon an anvil, by blows from a hammer. The last three metals in the table broke each at the first blow, with a clean and square fracture. The aluminum bronze only began to crack at the eighth blow, and required a number of additional blows before the two pieces were entirely separated. And the irregular, torn surface of the fracture showed the peculiarly tough and fibrous nature of the metal.

The elasticity of the aluminum bronze was tested by M. Tresca, Professor at the *Conservatoire des Arts et Métiers*. The experiment was made upon a bar of simple cast metal, and the following is his report: "The coefficient of elasticity of the aluminum bronze, the cast metal, is half that of the best

wrought-iron. This coefficient is double that of brass and four times that of gun metal, under the same conditions.”

The specific gravity is 7.7, about the same as iron. Another very valuable quality is presented in the fact that it is acted on by atmospheric influences less than are silver, brass, or bronze. This places it in the same rank with gold, platinum and aluminum.

Very stiff and very elastic, tougher than iron, very little acted upon chemically, and in certain cases not at all, capable of being cast like ordinary bronze or brass, forged like iron and steel, of being worked in every way like the most malleable metals or alloys, having, added to these properties, a color analogous to that of the most precious metal, this bronze proves itself adapted to uses almost innumerable. At first sight, it seems difficult to admit that the relatively small proportions of aluminum which enters into the composition of this bronze can be sufficient to modify so extraordinarily the properties of the copper which constitutes so large a portion of its weight. But we must remember that the specific gravity of aluminum is very low, and that a given weight of this metal possesses a bulk four times as large as the same weight in silver. It follows from this that the ten per cent. of aluminum contained in the bronze equals in bulk forty per cent. in silver.

The specimens of the ware we have seen, such as spoons, forks, cups, watch-cases, etc., are certainly very beautiful, having the color and high polish of gold, while dilute acids do not affect the surface.

On the Reduction of Silver in the Wet Way.

Every chemist is familiar with the reduction of chloride of silver in the form of powder by means of metallic zinc in the presence of a little free acid. It is not easy to bring two such substances as the silver salt and the metal into close contact, and after the work is accomplished the removal of the excess of zinc has its difficulties. Dr. Grager suggests a modification of the old method that ought to be more generally made known. The chloride of silver is dissolved in ammonia and poured into a well-stopped bottle, and into this is introduced an excess of metallic zinc, in not too small fragments, so that any reduced metal adhering to it may be readily washed off.

The decomposition begins immediately, and is rapidly accomplished, especially if the contents of the flask be well shaken up. Three hours will suffice to reduce one-quarter of a pound of chloride of silver. It is easy to ascertain when the reduction is ended, by testing a portion of the ammoniacal solution with hydrochloric acid. As soon as no cloudiness or curdy precipitate is formed, the work may be regarded as completed.

A slight excess of ammonia is said to be favorable. The reduced silver must be washed with water until all odor of ammonia has disappeared. The pieces of zinc are removed by pouring the contents of the flask through a funnel, the opening of which is too narrow for the passage of the zinc fragments, while the reduced silver can be easily washed through. The finely divided silver can be digested in hydrochloric acid to restore it to a pure white color, and it is then ready for solution or fusion, and will be found to be perfectly pure. In dealing with large quantities it would be economical to recover a portion of the ammonia by distillation. In the same way an ammoniacal solution of nitrate of silver can also be reduced by zinc, and the silver obtained pure, even when the original solution of the nitrate contains copper—provided a small quantity of silver be kept in the bath.

It is better where copper is present not to take all of the zinc that may be requisite for the reduction of the silver. It will prove a great convenience to be spared the necessity of converting the silver into the chloride, as it is no

easy task to wash out this salt on filters—and it will be found to be applicable to alloys which do not contain more than 25 per cent. of silver.—
From Prof. Joy in the Journal of Applied Chemistry.

Transcriber's Notes

Obvious errors in punctuation have been fixed.

[Page 7](#): “Mechanique Celeste” changed to “Méchanique Céleste”

[Page 12](#): “ou rexperience” changed to “our experience”

[Page 18](#): “head-quarters far astronomical observations” changed to “head-quarters for astronomical observations”

[Page 22](#): “it accomodates” changed to “it accommodates”

The [Table of Contents](#) lists “Equation of the Time Table” as the article on page 28. The actual article is named “On the Reduction of Silver in the Wet Way.” This has intentionally been left as per the original. Similarly, there is no actual section titled “Notices of New Tools” despite its inclusion in the Table of Contents, and this has been left as per the original.

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